Unit 1 : Introduction of Cloud Computing

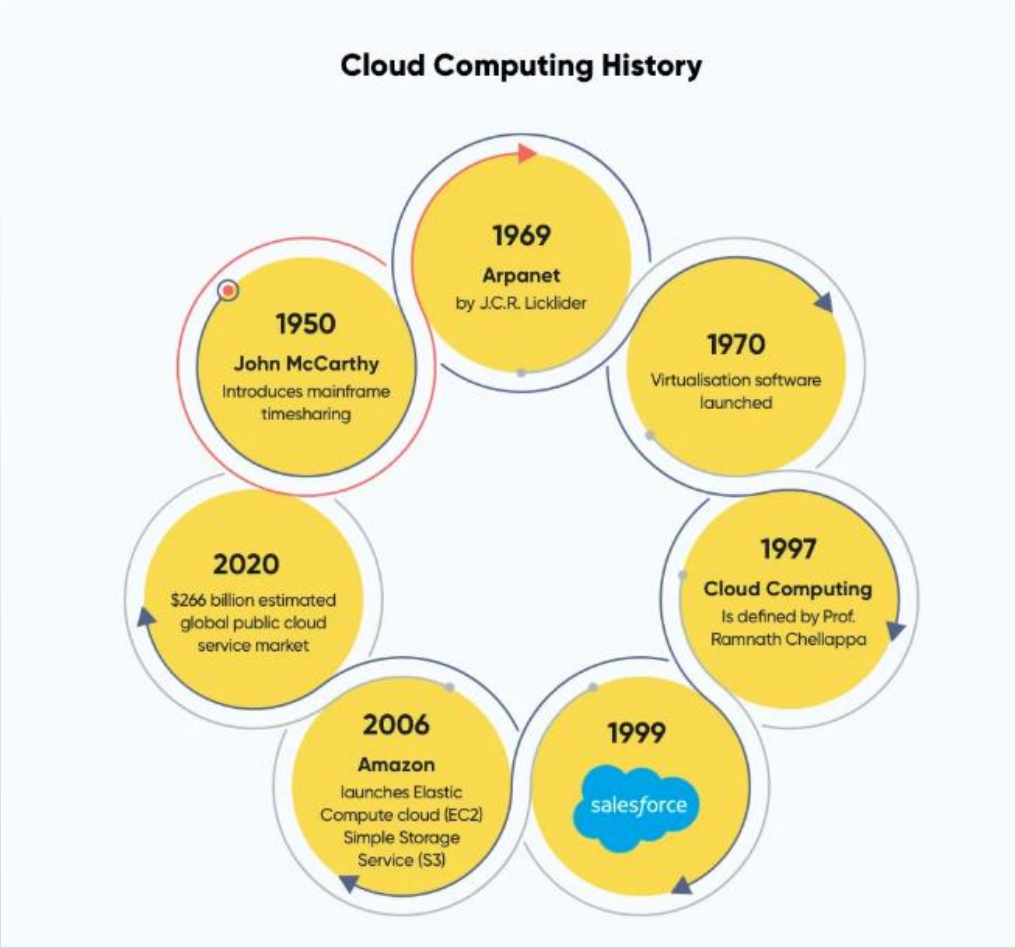
Topic

- ☞ Introduction Evolution
- ☞ Benefits and berries
- ☞ Cloud SPI Models
- ☞ Cloud Computing vs Cluster computing
- ☞ Cloud Architecture

Introduction

- Cloud Computing has fundamentally changed how businesses and individuals store, access, and manage data. It is a technology that allows you to use data and applications over the internet, moving beyond the limitations of a local computer's hard drive or a private server. But to truly understand its impact, we must first look at the problems it solved.
- **The Problem Cloud Computing Solves: The "Old Way" vs. The "New Way"**
- Before the cloud, if a company wanted to launch a website or an application, it had to follow a slow and expensive process:
- **The "Old Way" (On-Premises):** The company had to buy powerful physical servers, storage disks, and networking equipment. This required a huge upfront investment, known as **Capital Expenditure (CapEx)**. They had to guess their peak traffic needs, often buying far more capacity than they used day-to-day, leading to wasted resources. If they needed a new server, the process of ordering, installing, and configuring it could take weeks or even months.
- **The "New Way" (Cloud Computing):** The cloud changes this model entirely. Instead of buying hardware, you rent computing power from a cloud provider (like AWS, Google, or Microsoft). This shifts the cost from a large upfront investment to a manageable monthly bill, known as an **Operational Expenditure (OpEx)**. This model eliminates guesswork, long waiting times, and wasted resources.

Evolution


Cloud Computing History
1969
Arpanet
by J.C.R. Licklider
1970
Virtualisation software launched
1997
Cloud Computing
Is defined by Prof. Ramnath Chellappa
1999
salesforce
2006
Amazon
launches Elastic Compute cloud (EC2) Simple Storage Service (S3)
2020
$266 billion estimated global public cloud service market
1950
John McCarthy
Introduces mainframe timesharing

Evolution of Cloud Computing

- The phrase "Cloud Computing" was first introduced in the 1950s to describe internet-related services, and it evolved from distributed computing to the modern technology known as **cloud computing**.
- Cloud services include those provided by Amazon, Google, and Microsoft.
- Cloud computing allows users to access a wide range of services stored in the cloud or on the Internet.
- **Cloud computing** services include computer resources, data storage, apps, servers, development tools, and **networking protocols.**
- Cloud computing allows users to access a wide range of services stored in the cloud or on the Internet. **Cloud Computing** services include computer resources, data storage, apps, servers, development tools, and **networking protocols**.
- They are most commonly used by IT companies and for business purposes

1. Mainframe Computing(1950-1970)

- Mainframe which first came into existence in 1951 are highly powerful and reliable computing machines.
- These are responsible for handling large data such as massive input-output operations. Even today these are used for bulk processing tasks such as online transactions etc.
- These systems have almost no downtime with high fault tolerance. After distributed computing, these increased the processing capabilities of the system. But these were very expensive. To reduce this cost, cluster computing came as an alternative to mainframe technology.

2. Distributed Systems(1970-1980)

- The purpose of distributed systems is to share resources and also use them effectively and efficiently.
- Distributed systems possess characteristics such as scalability, concurrency, continuous availability, heterogeneity, and independence in failures.
- But the main problem with this system was that all the systems were required to be present at the same geographical location.
- Thus to solve this problem, distributed computing led to three more types of computing and they were-Mainframe computing, cluster computing, and grid computing.

3. Cluster Computing(1980-1990)

- In 1980s, cluster computing came as an alternative to mainframe computing.
- Each machine in the cluster was connected to each other by a network with high bandwidth.
- These were way cheaper than those mainframe systems. These were equally capable of high computations.
- Also, new nodes could easily be added to the cluster if it was required.
- Thus, the problem of the cost was solved to some extent but the problem related to geographical restrictions still pertained. To solve this, the concept of grid computing was introduced.

4. Grid Computing(1990-2000)

- In 1990s, the concept of grid computing was introduced.
- It means that different systems were placed at entirely different geographical locations and these all were connected via the internet.
- These systems belonged to different organizations and thus the grid consisted of heterogeneous nodes.
- Although it solved some problems but new problems emerged as the distance between the nodes increased.
- The main problem which was encountered was the low availability of high bandwidth connectivity and with it other network associated issues.
- Thus. cloud computing is often referred to as "Successor of grid computing".

5. Utility Computing(Late 1990-2000)

- Utility Computing is a computing model that defines service provisioning techniques for services such as compute services along with other major services such as storage, infrastructure, etc which are provisioned on a pay-per-use basis.

6. Virtualization(1980-Present)

- Virtualization was introduced nearly 40 years back.
- It refers to the process of creating a virtual layer over the hardware which allows the user to run multiple instances simultaneously on the hardware.
- It is a key technology used in cloud computing.
- It is the base on which major cloud computing services such as Amazon EC2, VMware vCloud, etc work on.
- Hardware virtualization is still one of the most common types of virtualization.

7. Web 2.0

- Web 2.0 is the interface through which the cloud computing services interact with the clients.
- It is because of Web 2.0 that we have interactive and dynamic web pages. It also increases flexibility among web pages.
- Popular examples of web 2.0 include Google Maps, Facebook, Twitter, etc. Needless to say, social media is possible because of this technology only. It gained major popularity in 2004.

8. Service Orientation

- A service orientation acts as a reference model for cloud computing.
- It supports low-cost, flexible, and evolvable applications. Two important concepts were introduced in this computing model.
- These were Quality of Service (QoS) which also includes the SLA (Service Level Agreement) and Software as a Service (SaaS).

Benefits and berries

- Emerging cloud computing technologies have created numerous challenges for handling data and information. Therefore, if you decide to implement cloud infrastructure services, here are some challenges of cloud computing and risks you can encounter.

No matter the size of the business, cloud computing advantages bring the potential to transform workloads, increasing efficiency and productivity.

Cost Savings

- Migrating to the cloud means less money spent up front on costly IT equipment. Likewise, technical benefits of cloud computing include the reduction of ongoing maintenance costs, IT personnel time, and rolling out upgrades.
- Instead, clients take advantage of the buying power and expertise of cloud providers. This leads to substantial cost savings – 80 per cent of IT teams saved money after moving to the cloud.

Security

- Data security is critical to business success, especially when storing sensitive information. Cloud services offer peace of mind through advanced security measures. They run state-of-the-art encryption, authentication, and security patches. Physical security is likewise improved, with data centres secured by several layers of protection.

Flexibility

- One of the most compelling benefits of cloud computing is its flexibility. Users select their providers, compute resources, storage space, and environments. This flexibility releases clients to create their ideal cloud setup.

Scalability

- Cloud environments are infinitely scalable. Companies can increase or decrease their bandwidth to cope with varying demands. Likewise, the scalability of cloud computing for businesses means cloud infrastructure grows seamlessly with your operation – never holding you back.

Rapid Deployment

- The rapid availability of cloud services gives users endless possibilities. New resources can be created and launched instantly, while developers can build and test without hardware constraints. This helps companies stay agile in a fast-moving world.

Mobility

- Cloud hosting allows users to access data from anywhere with an internet connection. In the age of smartphones, businesses can also offer their employees mobile solutions. Instead of being limited by local server access, companies can work from anywhere in the world – creating endless recruitment and growth opportunities.

Disaster Recovery

- Cloud environments can deploy across several data centre locations. If one server suffers downtime, services continue uninterrupted across other resources. Meanwhile, extensive cloud data storage options allow easy rollback of versions, backup recovery, and business continuity in the case of natural disasters.

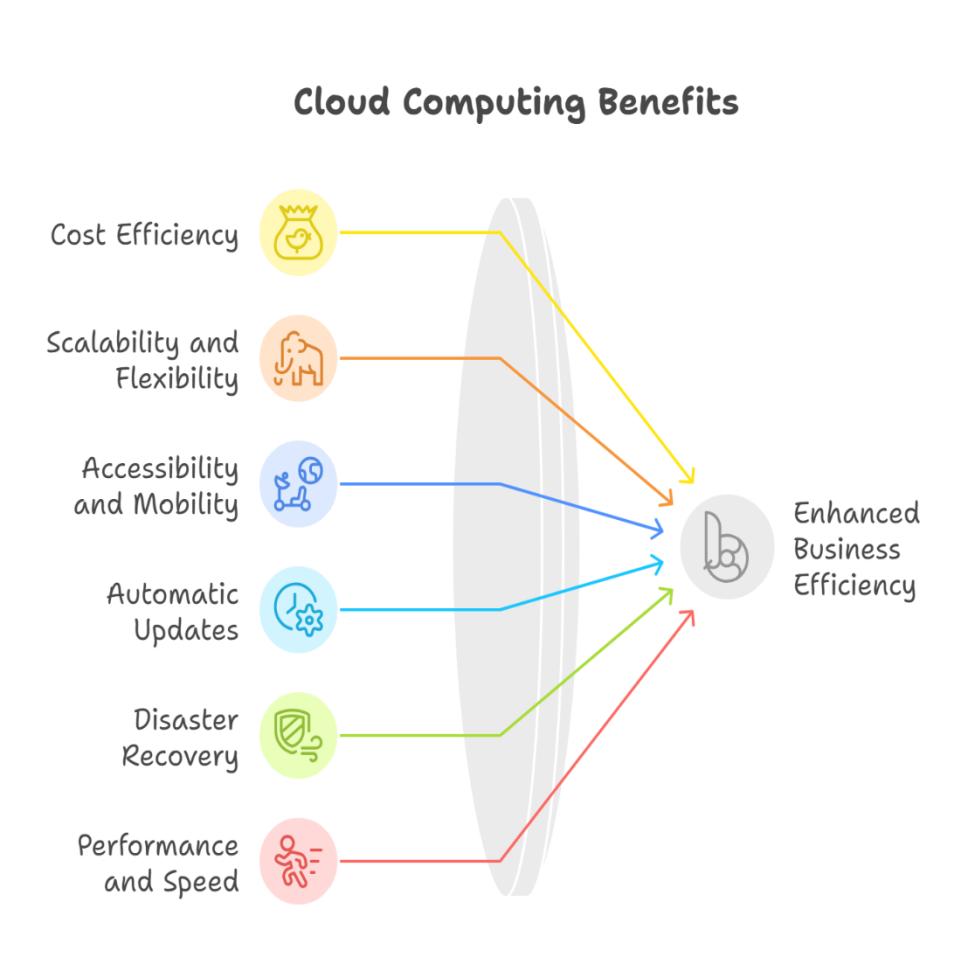
Reliability

- Downtime comes at a high cost – both financially and reputationally. Outages can drive customers away, cause lost revenue, and incur significant IT costs to repair. In a fully resilient multi-cloud environment, multiple resources help ensure business-critical uptime.

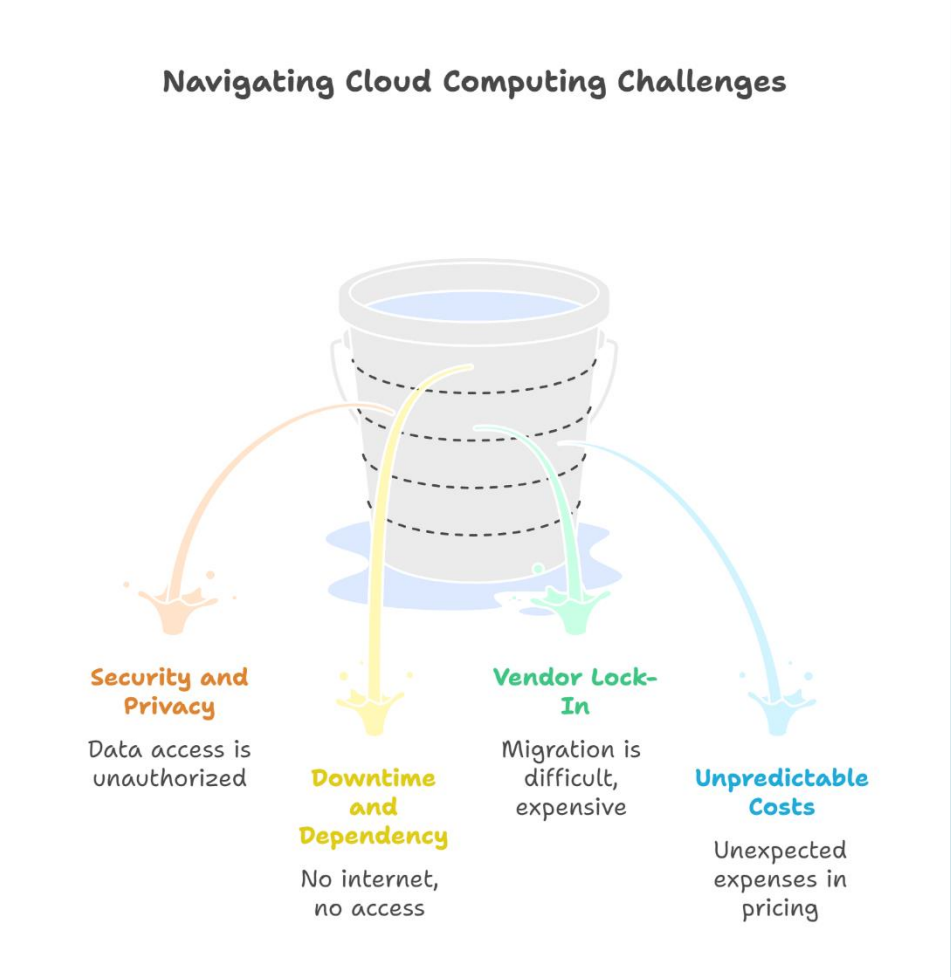
Collaboration

- With remote teams and increased mobility, collaboration is crucial. Cloud services allow users to work on tasks together, accessing on-demand software and resources to get the job done without being in the same location.

Benefits of Cloud Computing



Disadvantages of Cloud Computing



🚧 Barriers / Disadvantages of Cloud Computing

❌ 1. Security and Privacy Risks

- Despite advanced security features, data stored in the cloud may still be vulnerable.
- **Data Breaches:** Unauthorized access to sensitive business or customer data.
- **Loss of Control:** Data is stored on third-party servers, reducing control over security practices.
- **Shared Resources:** Multi-tenancy increases the risk of cross-tenant data leakage.
- **Compliance Issues:** Difficulty in meeting strict data protection laws (e.g., GDPR, HIPAA).

❌ 2. Downtime and Internet Dependency

- Cloud services rely heavily on stable internet connectivity and provider reliability.
- **Internet Outages:** No internet means no access to cloud data or apps.
- **Service Downtime:** Even top providers like AWS or Azure occasionally experience outages.
- **Latency Issues:** Delay in accessing data due to network congestion or server location.
- **Limited Offline Access:** Some cloud tools are not functional without an internet connection.

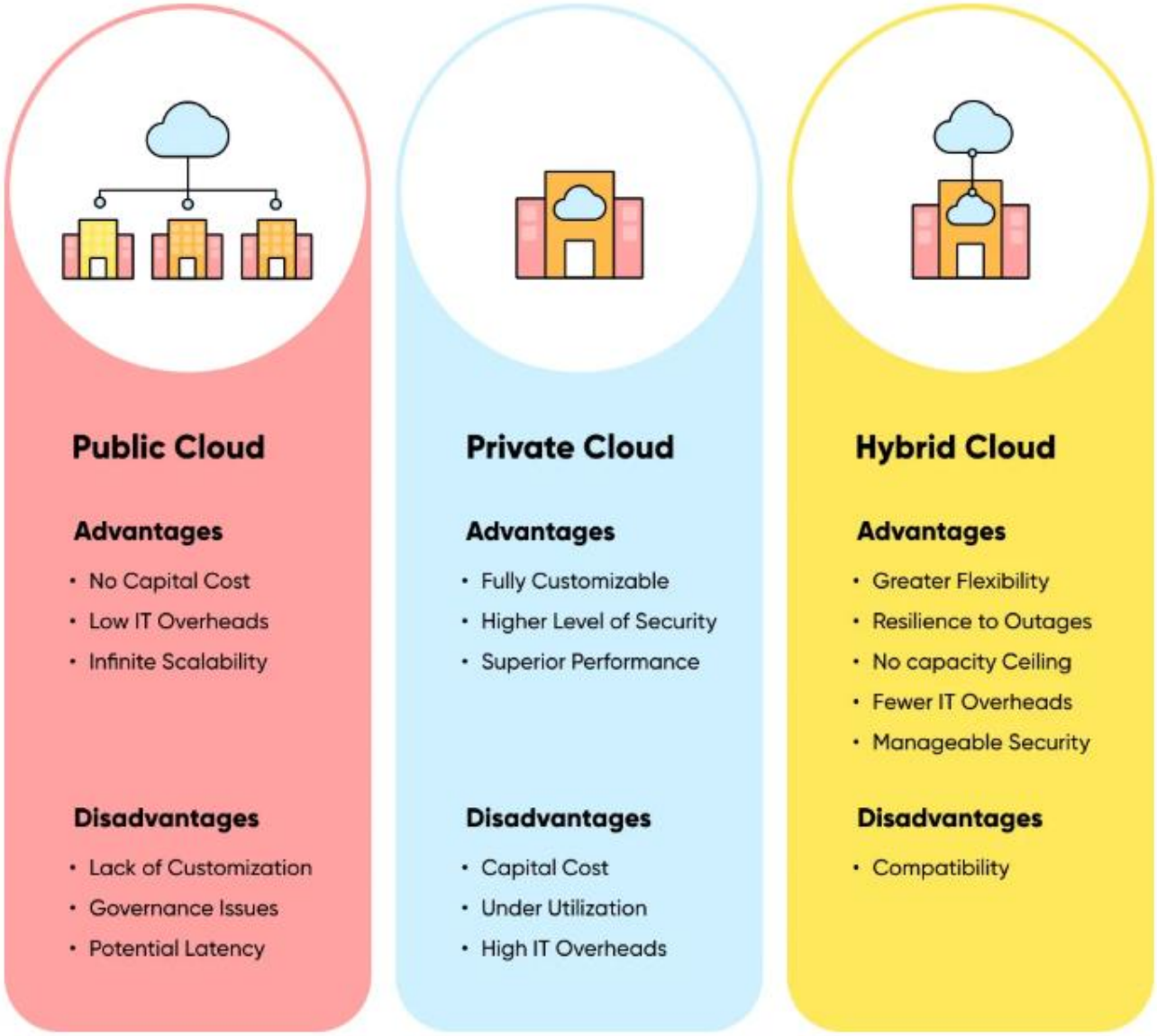
❌ 3. Vendor Lock-In

- Migrating from one cloud provider to another can be difficult and expensive.
- **Proprietary Technologies:** Cloud providers use unique tools and APIs that may not transfer easily.
- **High Switching Costs:** Data migration and retraining staff can be time-consuming and costly.
- **Compatibility Issues:** Applications built for one platform may not run on another without modifications.
- **Dependency on Single Provider:** Overreliance can be risky if the provider changes pricing, policy, or service availability.

❌ 4. Hidden and Unpredictable Costs

- Cloud pricing models can sometimes lead to unexpected expenses.
- **Pay-as-you-go Complexity:** Difficult to predict monthly bills with usage-based pricing.
- **Unmonitored Resource Use:** Leaving unused virtual machines or storage running can rack up costs.
- **Data Transfer Fees:** Moving data in/out of the cloud can incur additional charges.
- **Licensing Costs:** Some services or third-party apps require separate licenses not included in cloud packages.

Cloud SPI Models



1. Public Cloud

- Public clouds are operated and owned by third-party providers.
- In the case of a public cloud, the services offered by cloud providers are over a network that is open for public use.
- This implies that an organization shares the same hardware and network devices with other companies of the same cloud service provider.
- Microsoft Azure is an example of a public cloud.

2. Private Cloud

- A private cloud is a cloud deployment model that a single organization exclusively uses.
- In the case of a private cloud, the services and infrastructure are maintained on a private network, either located physically at the organization's on-site data center or hosted by third-party service providers.

3. Hybrid Cloud

- The combination of private and public clouds gives rise to a hybrid cloud.A hybrid cloud deployment model enables businesses to move applications and data between public and private clouds, leading to an agile, secure, and flexible digital business model.

Types of Cloud Services

1. Infrastructure as a Service (IaaS)

- Infrastructure as a Service (IaaS) is a cloud computing service where enterprises rent or lease servers for computing and storage in the cloud.
- Users can run any OS (operating system) or application on rented servers without worrying about the server's upkeep and running costs.
- IaaS provides compute storage, network, and load balancer services. IaaS provides shared infrastructure, web access to the resources, a pay-as-per-use model, and on-demand scalability.

2. Platform as a Service (PaaS)

- PaaS refers to an external cloud service provider providing and managing the hardware and an application-software platform.
- Still, the user is in charge of the apps that run on top of the platform, and the data such apps rely on.
- It makes it simple for programmers to build, test, use, and deploy web apps.
- PaaS provides different programming languages, application frameworks, databases, and other tools. PaaS provides simplified development, lower risks, pre-built business functionalities, and scalability.

3.Software as a Service (SaaS)

- SaaS is a service that offers consumers access to a managed software program hosted in the cloud.
- Most SaaS apps are web or mobile applications that consumers can access through a web browser.
- The user connects to the cloud applications through a dashboard or API, and software upgrades, bug fixes, and other general software maintenance are taken care of for them.
- SaaS provides business services, document management, social networks, and mail services.
- SaaS is easy to buy, has fewer hardware requirements, has low maintenance, multi-device support.

IaaS vs PaaS vs SaaS

Feature	Target User	Control Level	Example	User Manages	Provider Manages
IaaS	IT Admins, DevOps	High	AWS EC2, Azure VMs	OS, apps	Infra, virtualization
PaaS	Developers	Medium	Google App Engine, Heroku	Apps only	Infra + platform
SaaS	End-users	Low	Gmail, Salesforce	Nothing	Everything

Difference between Cloud Computing and Cluster Computing :

Serial Number	Category	Cloud Computing	Cluster Computing
1.	Goal	Providing on demand IT resources and services.	Performing a complex task in a modular approach.
2.	Resource Sharing	Specific assigned resources are not shareable.	Specific assigned resources are not shareable.
3.	Resource type	In cloud computing there is heterogeneous resource type.	In Cluster Computing there is homogeneous resource type.
4.	Virtualization	Virtualization hardware and software resources.	No virtualization resources.

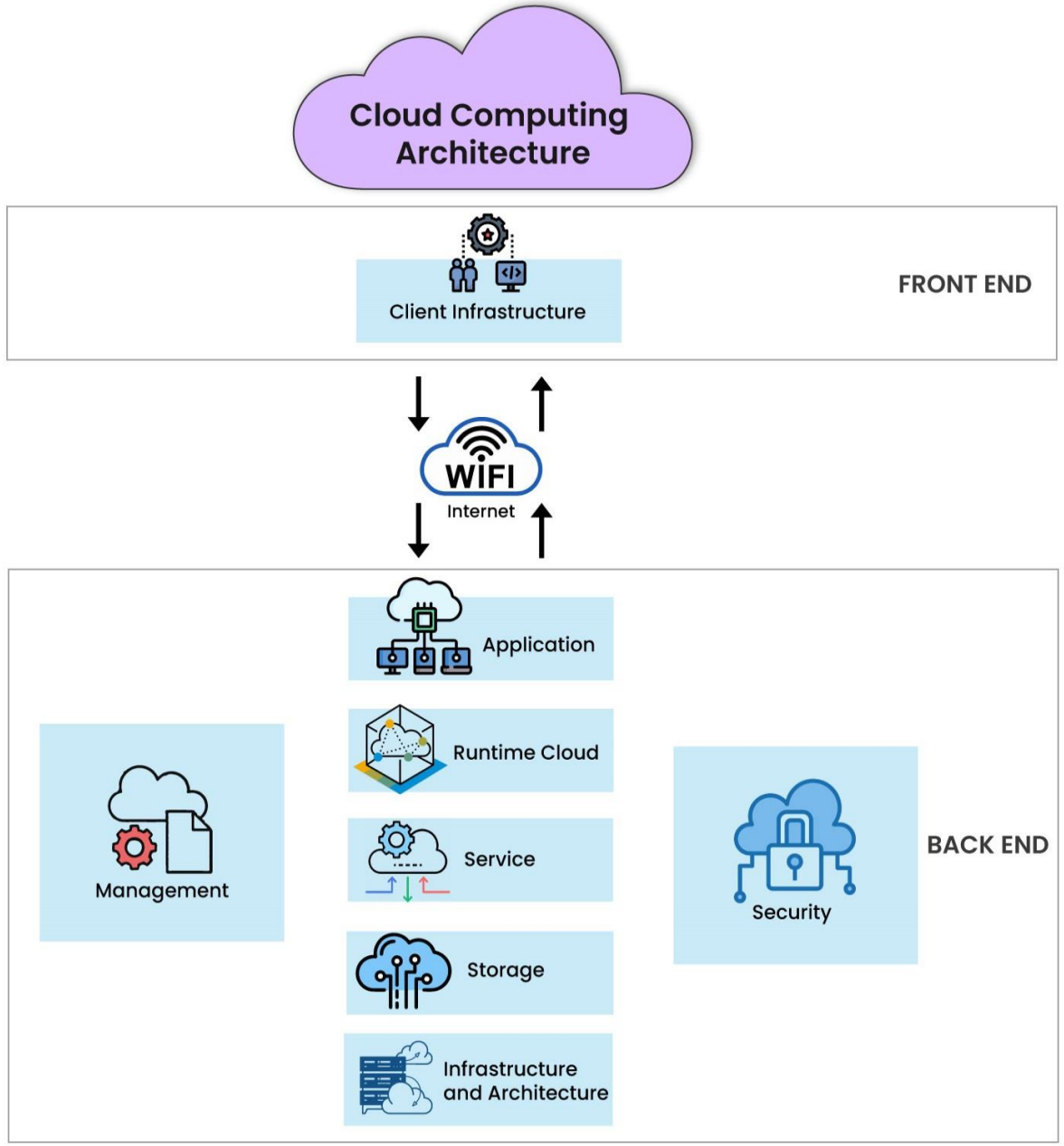
5.	Security	Security through isolation can be achieved.	Security through node credential can be achieved.
6.	Initial Cost	Initial capital cost for setup is very low.	Initial capital cost for setup is very high.
7.	Security Requirement	Very low	Very high
8.	Maintenance	Requires low maintenance.	Requires little more maintenance.
9.	Hardware	No hardware requirement physically.	More hardware requirement physically.
10.	Node OS	Multiple OS runs in VM	Windows, Linux

11.	User Management	User management is centralized or decentralized to vendor/third party.	User management is centralized.
12.	Scalability	Allowed	Limited
13.	Architecture	In Cloud Computing User chosen architecture.	In Cluster Computing Cluster oriented architecture
14.	Characteristic	Dynamic computing infrastructure/resources/services	Tightly coupled systems/resources
15.	Software Dependent	In cloud computing application domain independent software.	In cluster computing application domain dependent software.
16.	Example	Dropbox, Gmail	Sony PlayStation clusters

Cloud Architecture

- Cloud computing architecture is a hybrid design that combines Service Oriented Architecture (SOA) and Event-Driven Architecture (EDA).
- It refers to various components and subcomponents that work together to create the system's overall architecture.
- It offers higher bandwidth to its clients, due to which stored data in the cloud can be accessed from wherever they are in the world at any time.
- It allows businesses to lessen or eliminate their dependency on servers, storage, and networking infrastructure hosted on their own premises.
- The cloud architecture is divided into 2 parts, i.e.

1) Frontend
2) Backend


Cloud Computing Architecture
Client Infrastructure
FRONT END
WiFI
Internet
Application
Runtime Cloud
Service
Storage
Infrastructure and Architecture
Management
Security
BACK END

1. Frontend

- Frontend of the cloud architecture refers to the client side of cloud computing system.
- Means it contains all the user interfaces and applications which are used by the client to access the cloud computing services/resources.
- For example, use of a web browser to access the cloud platform.

2. Backend

- Backend refers to the cloud itself which is used by the service provider.
- It contains the resources as well as manages the resources and provides security mechanisms.
- Along with this, it includes huge storage, virtual applications, virtual machines, traffic control mechanisms, deployment models, etc.

Components of Cloud Computing Architecture

- **Client Infrastructure:** Client Infrastructure is a part of the frontend component.
- It contains the applications and user interfaces which are required to access the cloud platform. In other words, it provides a GUI(Graphical User Interface) to interact with the cloud.
- **Application :** Application is a part of backend component that refers to a software or platform to which client accesses. Means it provides the service in backend as per the client requirement.
- **Service:** Service in backend refers to the major three types of cloud based services like SaaS, PaaS and IaaS. Also manages which type of service the user accesses.
- **Runtime Cloud:** Runtime cloud in backend provides the execution and Runtime platform/environment to the Virtual machine.
- **Storage:** Storage in backend provides flexible and scalable storage service and management of stored data.
- **Infrastructure:** Cloud Infrastructure in backend refers to the hardware and software components of cloud like it includes servers, storage, network devices, virtualization software etc.

- **Management:** Management in backend refers to management of backend components like application, service, runtime cloud, storage, infrastructure, and other security mechanisms etc.
- **Security:** Security in backend refers to implementation of different security mechanisms in the backend for secure cloud resources, systems, files, and infrastructure to end-users.
- **Internet:** Internet connection acts as the medium or a bridge between frontend and backend and establishes the interaction and communication between frontend and backend.
- **Database:** Database in backend refers to provide database for storing structured data, such as SQL and NOSQL databases. Example of Databases services include Amazon RDS, Microsoft Azure SQL database and Google CLoud SQL.

Cloud Computing Architecture Example

1. Online Learning App (GeeksforGeeks Classroom)

- Students are using GeeksforGeeks application for watching tutorials for learning purpose, they solve different coding problems and take quizzes also for testing their skills.
- For storing videos and study materials Geeksforgeeks are using cloud services like Amazon S3.
- It also uses AWS Lambda to run small backend tasks like checking quiz answers or updating progress.
- For managing users and logins securely, services like AWS IAM (Identity and Access Management) are used

- **2. Online Store (E-Commerce Website)**
- Everyone is using online websites like Amazon or Flipkart for online shopping.
- It uses cloud computing to run its website and manage orders.
- When you browse and buy products, you're using the front end. In the background, cloud services keep track of what's in your cart, handle payments, and update stock.
- The product photos are saved in cloud storage like Amazon S3, and customer details are stored in databases.
- Cloud tools also watch over the site to keep it secure and running fast.

- **3. Mobile App Backend (Food Delivery App)**
- Think of a food delivery app like Zomato or Swiggy. You use the app to order food, and everything works smoothly just because of cloud computing.
- The app you see on your phone is the front layer. When you order something, cloud services handle the process in the background like telling the restaurant, tracking your order, and handling payments.
- The servers that run the app live in cloud data centers like Google Cloud or AWS, and all your data is kept safe using secure login systems.

Benefits of Cloud Architecture

- **Scalability**
- **High Availability**
- **Disaster Recovery**
- **Security**
- **Reduced Costs**
- **Faster Time-to-Market**